

# Does Hybridisation Pay? A Systematic Review of Metaheuristic Optimisation and Machine Learning Hybrids in Pavement Engineering (2005–2026)

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## Abstract

PLACEHOLDER — structured abstract, ~280 words, written last so that every number in it traces to the audit. Do not draft this before Section 9 is final.

*Keywords:* pavement engineering, metaheuristic optimisation, machine learning, hybrid models, systematic review, methodological rigour

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**Draft status.** This file renders to PDF, Word and HTML from one source. Sections marked *PLACEHOLDER* are not yet written. Every citation resolves against `references.bib`, which is generated from `data/seed_bibliography.csv`; a citation key that is not in the verified database fails the render rather than passing silently.

## 1. Introduction

PLACEHOLDER — see `docs/01_SCOPE_AND_TAXONOMY.md` §1 for the argument this section must carry: hybridisation is the field’s dominant claim of novelty, and the marginal benefit of hybridisation over a properly tuned base learner is almost never tested.

The review literature this study builds on is summarised in Figure 1. Prior surveys catalogue where neural and hybrid methods have been applied [1, 2, 3]; two recent contributions name the underlying problem without converting it into per-study evidence [4, 5].

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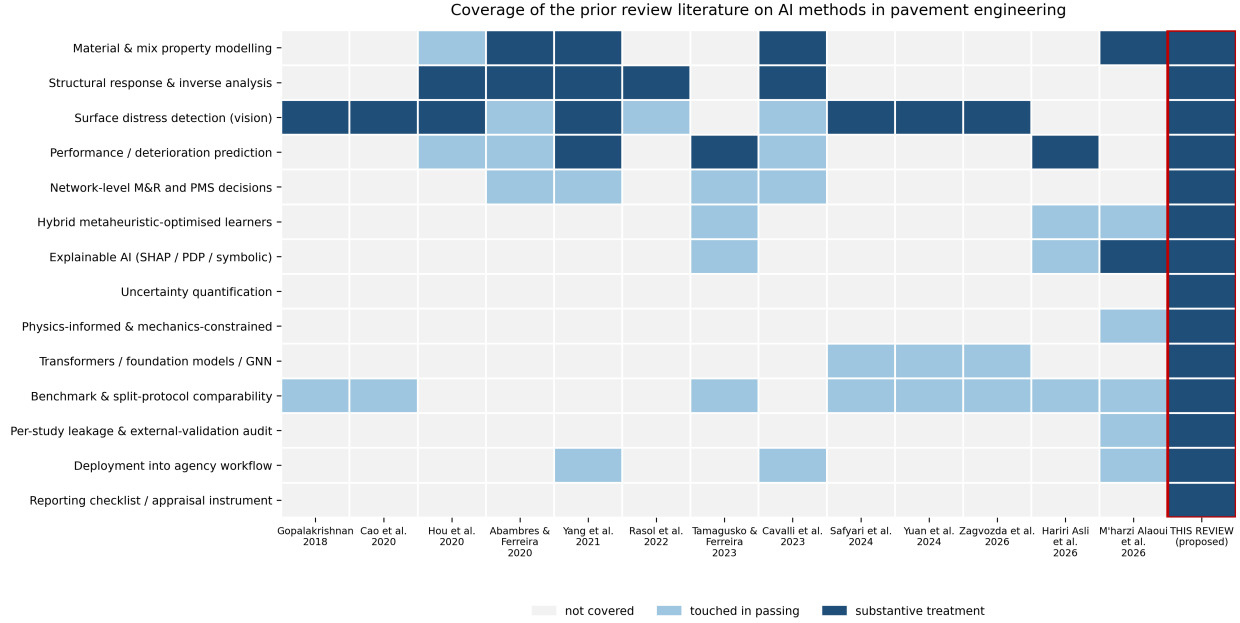


Figure 1: Coverage of the prior review literature on artificial-intelligence methods in pavement engineering. Scoring: 2 = substantive dedicated treatment, 1 = touched in passing, 0 = absent.

## 2. Review methodology

*Full draft in docs/02\_MANUSCRIPT\_S2\_methodology.md; transferred here section by section as each is finalised.*

### 2.1. Scope and framing question

PLACEHOLDER — transfer from the draft.

### 2.2. Information sources and search

Records were identified from OpenAlex, Scopus, Web of Science Core Collection and Semantic Scholar, with backward and forward snowballing from the 13 prior reviews listed in Table 1. The window runs from 1 January 2005 to 30 June 2026.

Table 1: Prior reviews of artificial-intelligence methods in pavement engineering, with the territory each occupies.

| Review         | Year | Venue       | Territory          |
|----------------|------|-------------|--------------------|
| Gopalakrishnan | 2018 | Data        | distress-detection |
| Cao et al.     | 2020 | IEEE Access | distress-detection |

Table 1: Prior reviews of artificial-intelligence methods in pavement engineering, with the territory each occupies.

| Review                | Year | Venue                               | Territory                                |
|-----------------------|------|-------------------------------------|------------------------------------------|
| Hou et al.            | 2020 | Engineering                         | structural-evaluation/distress-detection |
| Abambres et al.       | 2020 | TechRxiv (preprint)                 | all                                      |
| Yang et al.           | 2021 | J. Traffic Transp. Eng. (Engl. Ed.) | all                                      |
| Rasol et al.          | 2022 | Constr. Build. Mater.               | structural-evaluation                    |
| Cavalli et al.        | 2023 | J. Road Eng.                        | all                                      |
| Tamagusko et al.      | 2023 | Infrastructures                     | performance-prediction                   |
| Yuan et al.           | 2024 | Remote Sensing                      | distress-detection                       |
| Safyari et al.        | 2024 | Sensors                             | distress-detection                       |
| M'harzi Alaoui et al. | 2026 | J. Umm Al-Qura Univ. Eng. Archit.   | materials                                |
| Zagvozda et al.       | 2026 | Adv. Civ. Archit. Eng.              | distress-detection                       |
| Hariri Asli et al.    | 2026 | J. Infrastruct. Preserv. Resil.     | performance-prediction                   |

### 2.3. Eligibility

PLACEHOLDER — transfer from the draft, including the structural (not lexical) definition of hybridity in docs/01\_SCOPE\_AND\_TAXONOMY.md §2.

## 3. Bibliometric landscape

PLACEHOLDER — written after the harvest completes.

## 4. Taxonomy of hybridisation

PLACEHOLDER — the H1–H7 taxonomy with theoretical justification and characteristic failure mode, from docs/01\_SCOPE\_AND\_TAXONOMY.md §3.

## 5. Hybrid models in material and mix modelling

PLACEHOLDER.

## 6. Hybrid models in structural evaluation and inverse analysis

PLACEHOLDER — anchored on back-calculation [6? , 7].

## 7. Hybrid models in performance prediction

PLACEHOLDER.

## 8. Physics-informed, fusion, stacking, decomposition and symbolic hybrids

PLACEHOLDER — [8] for the physics-informed case; [9] for the symbolic-numeric case.

## 9. The hybridisation premium

PLACEHOLDER — the paper’s central analysis. Definition and the evidence already in hand are in docs/01\_SCOPE\_AND\_TAXONOMY.md §4 [10, 11, 12].

## 10. Interpretability and uncertainty in hybrid pipelines

PLACEHOLDER.

## 11. From premium to decision

PLACEHOLDER — built on the cost-of-prediction-error curve in [13].

## 12. PAVE-ML: a reporting and appraisal checklist

PLACEHOLDER — instrument in docs/03\_PAVE-ML\_instrument.md.

## 13. Research agenda

PLACEHOLDER.

## 14. Conclusions

PLACEHOLDER.

## Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

## Declaration of generative AI in the writing process

PLACEHOLDER — to be completed per the target journal’s wording before submission. Elsevier requires a statement naming the tool, the purpose of use, and confirmation that the authors reviewed and take full responsibility for the content. Disclosed use is permitted; undisclosed use is not.

## Data availability

The screening database, coding rubric, harvesting scripts and the full analysis pipeline that produced every figure and table in this paper are available at [REPOSITORY URL].

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